

SOLAR PV CELLS TARGET GRID PARITY

The main challenge before solar cell manufacturers is to reduce the cost of devices through increase in efficiency, reduction of rejection and increase in volume of production



Solar cell line of Solar Semiconductor

■ UMA BANSAL

The solar photovoltaic (PV) manufacturing base in India comprises primarily module manufacturers and now an increasing number of cell manufacturers. During 2008-09, 15 companies were actively engaged in manufacture of solar cells and 20 in manufacture of PV modules. Around 50 companies were engaged in manufacturing a variety of PV systems.

Compared to the global scale, the

Indian market has so far remained very small. And leading Indian solar PV cells and modules manufacturing companies have been exporting most of their production to established markets such as Germany, Spain and the US.

"Production capacity in India is currently estimated in excess of 400 MW for cells and 1000 MW for modules, and is expected to cross 750 MW and 1250 MW, respectively, by the end of year 2010," informs Sathya Prasad, president, SEMI India.

These capacities are currently adequate to take care of the requirements of the first phase of the National Solar Mission, which targets to install 1000 MW of grid-connected solar plants by March 2013. Eventually, the Mission targets to have up to 5000 MW of solar manufacturing capacity by 2016-17.

Most of the demand in India is for domestic home lighting systems for rural areas, where there is no light and the government can't lay transmission lines. These are decentralised systems,